Report on SHMM-2_fCL

Semi Hidden Markov Model, 2nd order, for front-month WTI Crude Oil futures

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1. Abstract

The present model is part of a broader research series attempting to model forward price behavior of NYMEX WTI Crude Oil futures (CL).

It was designed as a successor to HMM-fCL (formerly the CLM26-Regime-Lab) and MCwNNrm-fCL addressing the memoryless limitations of a first-order HIdden Markov Model and the Monte Carlo Nearest Neighbors inability to distinguish economically distinct tail-event catalysts. The proposed architecture is a duration, conviction, and hazard-aware, second-order semi-Hidden Markov Model capable of modeling regime persistence, flip timing, and directional transitions with explicit-run length dependence.

A leakage-controlled backtest and evaluation harness was used to assess next-state accuracy, flip-only regime accuracy, and directional-probability calibration. The model demonstrates strong structural coherence, high next-state accuracy, good next-regime accuracy, and stable duration-conditioned behavior. SHMM-2_fCL is now the canonical model for this research line and the foundation of future improvements.

2. Background and Motivation

Modeling the 24-hour forward return behavior of front-month WTI Crude Oil (CL) requires accounting for two defining characteristics of the asset: its regime-dependent volatility and drift, and divergent catalyst-driven shock tail behavior. The first model in this research series, HMM-fCL, successfully captured low-volatility return patterns but was limited by the memoryless nature of a first-order Hidden Markov Model. Without explicit duration structure, the model could not distinguish between temporary or transitionary regimes and continuous regimes. In addition, the model was unable to produce discrete, directionally calibrated forecasts.

The subsequent model, MCwNNrm-fCL, attempted to address these limitations by combining a Monte Carlo baseline with a Nearest-Neighbor residual modifier. Despite extensive experimentation, the hybrid architecture demonstrated no persistent predictive edge. Its failure was structural: Monte Carlo distributions could not represent the heavy-tailed, catalyst-driven nature of CL returns, and local pattern similarity could not meaningfully correct for exogenous shocks.

These limitations motivated a pivot toward a structural, duration-aware regime model. SHMM-2_fCL was designed to explicitly model regime persistence, run-length dynamics, and second-order transitions, while incorporating conviction-based hazard deformation and a new direction-confidence scoring system. The goal was to build a model aligned with the true structure of CL returns rather than attempting to correct a mis-specified baseline

3. Model Architecture

The SHMM-2_fCL engine is a second-order Semi-Hidden Markov Model with explicit duration dependence, conviction-aware hazard deformations, and a five-regime topology. The architecture was designed to generate a discrete next state and next regime predictions for front-month WTI Crude Oil (CL) while remaining fully leakage-controlled and reproducible.

Data

This analysis uses 1-hour returns from the WTI Crude Oil continuous futures contract (CL), E-mini S&P 500 continuous futures contract (ES), E-mini Nasdaq-100 continuous futures contract (NQ), COMEX gold futures (GC), and two FX contracts (EU and JY) over the period September 27, 2009 to present. This timeframe is in line with other studies in this series. Data was obtained from commercial historical market data vendor Kibot, containing timestamp-aligned intraday futures data with full 24-hour Globex session coverage and the standard OHLCV fields (open, high, low, close, and traded volume).

3.1 Second-Order Transition Structure

The base transition object is a second-order tensor, defined by:

$$T[i, j, k] = P(S_t = k \mid S_{t-1} = j, S_{t-2} = i)$$

This captures directional momentum, volatility clustering, and catalyst-driven discontinuities more effectively than the first-order model.

The model also expands the regime count to five:

- 0: low-vol up
- 1: low-vol down

- 2: high-vol up
- 3: high-vol down
- 4: chop

The addition of the chop regime isolates drift-neutral, volatility-compressed periods that were previously forced into a directional regime.

3.2 Scalar Encoding of the Previous Regime

A key innovation is treating the previous regime as a scalar feature, not a categorical index. This enables a continuous interpolation across regime identities, smoother hazard deformation, more expressive duration-conditioned transitions, and a unified representation of regime identity and regime age. This scalar representation is used directly in the next-state and next-regime equations.

3.3 Hazard Deformation Pipeline

The raw transition slice $T\,[\,i,\,j,\,:\,]$ is modified through a deterministic pipeline utilizing global persistence scaling, age-dependent self-transition deformation, regime-specific hazard multipliers, conviction smoothing, a conviction-age interaction multiplier, and a normalization process. This pipeline produces a hazard-adjusted transition vector that reflects both continuous drift and discrete catalysts.

3.4 Unifying Formula: Next-state Probability

The next-state distribution is defined as:

$$P\,(\,S_{t+1}\;=\;k\,)\;=\;\frac{T\,[\,s_{t-1},\,s_t,\,k\,]\cdot H\,(\,s_t,\,age_t\,)\cdot C\,(\,conv_t,\,age_t\,)}{\sum_r T\,[\,s_{t-1},\,s_t,\,r\,]\cdot H\,(\,s_t,\,age_t\,)\cdot C\,(\,conv_t,\,age_t\,)}$$

Where $T\,[\,s_{t-1},\ s_t,\ k\,]$ is the second-order transition tensor, $H\,(\,s_t,\ age_t\,)$ is the hazard deformation applied to the self-transition, and $C\,(\,conv_t,\ age_t\,)$ is the conviction-age interaction multiplier. This formula defines the full next-state probability vector, which is used for next-state accuracy, next-regime probability, direction-confidence probability, and the time to next-flip estimation.

3.5 Unifying Formula: Next-Regime Probability

The model predicts the next regime by conditioning on the probability of leaving the current regime which is defined as:

$$P\,(\,flip\ at\ t\ +\ 1\,)\ =\ 1\ -\ P\,(\,S_{t+1}\ =\ s_t\,)$$

The conditional probability of flipping to regime *k* is defined as:

$$P\,(\,S_{t+1}\ =\ k\,|\,S_{t+1}\ \neq\ s_t\,)\ =\ \frac{P\,(\,S_{t+1}=k\,)}{\sum_{r\neq s_t} P\,(\,S_{t+1}=r\,)}$$

The predicted flip destination is:

$$\widehat{s_{t+1}^{flip}}\ =\ argmax_{k\neq s_t}\,P\,(\,S_{t+1}\ =\ k\,)$$

This formulation separates flip timing (a hazard driven value) from flip destination (transition-driven value), allowing the model to produce coherent-run level predictions at higher accuracy.

3.6 Directional-Confidence Probability

The model aggregates probability mass across directionally congruent regimes defining directional confidence probability as:

$$dirprob_t = \sum_{r : dir(r) = \widehat{dir(s_{t+1})}} P(S_{t+1} = r)$$

Clustering directional groups as follows:

- Positive cluster: {0, 2}
- Negative cluster: {1, 3}
- Neutral cluster: {4}

This provides a high-level directional probability metric that is more stable than regime-level predictions.

4. Limitations, Findings & Discussion

SHMM-2_fCL resolves the primary structural flaws of both HMM-fCL and MCwNNrm-fCL, despite this there are still structural limitations worth addressing. This section covers those concerns and provides empirical analysis of the model.

4.1 Structural Limitations

Arbitrary Regime Assignments

The core four directional regimes (low-vol up, low-vol down, high-vol up, high-vol down) remain hand-assigned abstractions of true microstructure behavior rather than empirically discovered latent states. While the addition of the chop regime improves separation between drift-neutral and directional periods, the directional regimes themselves still compress a wide range of geometric clusters into coarse groups. This limits the model's ability to distinguish continuous catalysts vs. discrete catalysts, microshock reversion vs. persistent directional runs, and volatility-driven transitions vs. drift-driven transitions. Geometrically defined regimes could help to alleviate this issue, though a lack of directional bias in them may lower overall model utility.

Shock Responsiveness and Volatility Calendars

Although SHMM-2_fCL is significantly more shock-aware than its predecessors, it still relies on fixed-window volatility calendars. This creates two issues:

- The model can react too slowly to sudden volatility expansions, especially when the shock occurs in a regime run when the age persistence multiplier is the strongest.
- The hazard deformation pipeline can only partially compensate for this.

While manual overrides correct this second issue, the model alone may lag during extreme events. A variable-window or adaptive volatility calendar would improve responsiveness, and alleviate this issue.

Lack of a Rolling Timeframe

The model is trained on the full historical dataset it was fed, which includes microstructure regimes that may no longer exist. Because SHMM-2_fCL does not use a rolling timeframe, older data can distort transition probabilities, inflate persistence in outdated regimes, or bias hazard curves toward historical microstructure patterns. This is the most significant remaining structural limitation. A rolling-window or exponentially decayed calibration would mitigate this issue and is a priority for future iterations.

4.2 Empirical Findings

Next-State Accuracy

The model achieves a next-state accuracy of 0.8754, at the time of this paper's creation. This metric however is not inherently meaningful, instead being largely an artifact of strong regime persistence in CL and the second-order structure reinforcing directional continuity. The model is very good at predicting that the current regime will continue, but this is expected given the underlying data-generating process.

Next-Regime Accuracy

The model achieves a next-regime accuracy of 0.5561, at the time of this paper’s creation. This metric evaluates the model only on periods where the true regime changes, making it a direct measurement of the models ability to anticipate regime transitions, identify the correct destination regime, and distinguish between directional and volatility-driven flips. A random

baseline approximation would be 0.2500 (four possible non-self regimes), contextualizing 0.5561 as a meaningful predictive edge and a statistically significant improvement over chance.

This model demonstrates true alpha in regime-transition prediction.

4.3 Tradeability and Practical Implications

The model's structure makes it inherently tradeable providing explicit next-state and next-regime, enter-exit logic based on predicted remaining time in regime, and safety filters based upon chop or extreme-volatility period. Ultimately the model provides stable and interpretable signals that align with the underlying market microstructure, offering a tradable basis for future extensions.

4.4 Comparative Discussion

SHMM-2_fCL represents a structural breakthrough relative to prior models resolving the memoryless static prediction problem of HMM-fCL by introducing explicit duration dependence and second-order transitions, resolving the catalyst homogeny problem of MCwNNrm-fCL by modeling regime persistence and flip timing structurally rather than relying on local pattern similarity while also allowing for manual user input, and by producing the first statistically meaningful next-regime accuracy in the research series.

While limitations in the system still remain, specifically referencing the lack of a rolling timeframe and the current classification structure, the model provides a coherent, structurally aligned foundation for both research and market application.

5. Conclusion

SHMM-2_fCL marks a structural turning point in this research series. By replacing distribution-based and pattern-matching approaches with a duration-aware, conviction-interpreting, second-order regime engine, the model resolves the two core failures of its predecessors: the memoryless static prediction of HMM-fCL and the catalyst homogeny and structural discontinuity of MCwNNrm-fCL. The five-regime topology, scalar previous-regime encoding, and unified next-state next-regime formulations provide a coherent framework for modeling regime persistence, flip timing, and directional transitions in front-month WTI crude oil.

While limitations remain, the empirical results demonstrate non-trivial alpha. The next-state accuracy reflects the persistence of CL regimes, and the next-regime accuracy on flips represents a genuine edge relative to a random baseline. Combined with stable enter/exit logic and safety filters, SHMM-2_fCL is the first model in this line that is both structurally aligned with the underlying market and practically tradable.